HOUSE PRICE PREDICTION – PROJECT SUMMARY

This project focused on predicting house prices using machine learning techniques and analyzing the factors that influence property value. The Housing dataset was loaded, explored, cleaned, and preprocessed using Python libraries such as Pandas and Scikit-learn.

Data preprocessing included handling missing values, removing duplicate records, and converting categorical variables into numerical form through one-hot encoding. The cleaned dataset was then divided into training and testing sets using an 80:20 ratio.

Two regression models were developed and evaluated: Linear Regression and Random Forest Regressor. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Linear Regression achieved the best overall performance with an R^2 Score of approximately 0.65, indicating that the model explained around 65% of the variation in house prices.

Data visualization techniques were used to gain additional insights. A histogram showed that house prices follow a right-skewed distribution, while the correlation heatmap identified area, bathrooms, stories, parking, and air conditioning as key factors affecting house prices. An Actual vs Predicted scatter plot demonstrated that the model was able to capture the overall trend of property values reasonably well.

The analysis suggests that larger houses with better facilities and amenities generally command higher prices. These findings can help real estate businesses make more informed pricing decisions and better understand market trends.